Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/17/2021
PI Name:	Karma, Alain Ph.D.		
Project Title:	Computational Studies of Microstructure Formation during Alloy Solidification in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	NOTE: Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX16AB54G," with the same Principal Investigator, Dr. Alain Karma. Dendritic microstructures formed during solidification processing (such as casting, welding, and brazing) determine the properties of structural metallic alloys used in aerospace materials essential for NASA exploration in space and produced in numerous US industries with commercial applications ranging from automobile engine parts to turbine blades used in aircraft and power generation. The main project goal is to develop multiscale and multiphysics computational models to relate solidification processing conditions with alloy microstructures and properties. Those models provide enhanced capabilities to accelerate the development of advanced (light-weight, high-temperature, and extreme environment) aerospace materials produced on ground by diverse solidification processes including emerging		

Task Description:	new technologies such as additive manufacturing. In addition, they provide support for materials processing on extraterrestrial surfaces in a reduced gravity environment. Model development is carried out iteratively with microgravity solidification experiments carried out in the Materials Science Laboratory and Transparent Alloy and Device for the Study of Critical Liquids and Crystallization (DECLIC) facilities aboard the International Space Station (ISS) in collaboration with ESA (European Space Agency) and CNES (Centre national d'études spatiales, French space agency) research teams. Computationally-designed materials has been identified as a critical technology in the NASA Technology Roadmap. This research meets the general objective of this critical area of accelerating materials development by integrating physics-based computational models of materials at multiple length scales to relate solidification processing, microstructures, and properties, and by developing massively parallel implementation of computational codes on graphics processing units (computer power). The present project makes use of both phase-field (PF) simulations and the dendritic needle network (DNN) model to establish benchmark comparisons between theory and microgravity experiments. Phase-field simulations resolve the details of the dendrite tip scale, while the DNN model tracks the dynamics of dendritic networks on a much larger scale of several grains. For this reason, the DNN model has emerged as a powerful method to bridge the dendrite tip and grain scales. Those simulation tools provide the means to model microstructure formation in a diffusive growth regime over the full range of scales needed to establish quantitative benchmark comparisons with microgravity observations. Ongoing computational studies have two main objectives: DECLIC/DSI. The first objective is to model directional solidification experiments in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) onboard the ISS as part of a CNES/NASA project. Those experiments visualize for the first time in dilute transparent alloys the solid-liquid interface evolution in three dimensions under diffusive growth conditions. They provide a wealth of new observations to elucidate complex mechanisms governing the spatiotemporal evolution of three-dimensional dendritic array structures. The goal of this research is to model both DECLIC/DSI experiments and the new series of DECLIC/DSI-reflight(R) experiments. CETSOL (Columnar-to-Equiaxed Transition in Solidification Processing (CETSOL). The second objective is to model directional solidification experiments in metallic and concentrated transparent organic alloys in the framework of the ESA/NASA CETSOL-6 project (March 2018 – February 2021) carried out in collaboration with US and European partners. The general goal of this project is to improve the understanding of complex physical phenomena governing the Columnar-to-Equiaxed Transition (CET) in alloy SOLidification and its consequences for industrial castings. This project involves studies of metallic alloys and the transparent alloy system Neopentylglycol-(d)Camphor (NPG-DC) in the Transparent Alloy facility. This system adds the capability to visualize both columnar and equiaxed dendritic grain development and the CET in a concentrated alloy regime, thereby complementing the DECLIC/DSI-R studies of columnar dendrites in dilute SCN-camphor alloys.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to lightweight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales. Those tools provide enhanced capabilities to accelerate the development of advanced (light-weight, high-temperature, and extreme environment) aerospace materials produced on ground by diverse solidification processes including emerging new technologies such as additive manufacturing. In addition, they provide support for materials processing on extraterrestrial surfaces in a reduced gravity environment.
Task Progress:	The major accomplishments during the present grant period include: • A detailed quantitative computational study of the growth competition of columnar grains of different crystalline orientations in a well-developed dendritic regime characteristic of directional solidification experiments of metallic alloys conducted in the International Space Station (ISS). Dendritic microstructures are highly branched and hierarchically organized in primary, secondary, and tertiary branches. By performing an extensive series of simulations of the directional solidification of binary alloys, we were able to pinpoint for the first time a key microstructural length scale that plays a dominant role in controlling the orientation of the boundary between columnar grains as a function of the temperature gradient and the crystal orientations of two neighboring grains (bi-crystallography). The results identify this length scale as the spacing between secondary dendrite branches preceding the emergence of tertiary branches. The identification of this length scale makes it possible to carry out larger-scale cellular automaton simulations of polycrystalline solidification, thereby significantly enhancing the computational capability to predict dendritic alloy microstructures in space and on Earth. • A phase-field study of the effect of a radial temperature gradient on microstructure evolution during directional solidification of binary alloys. While solidification fronts with a macroscopic curvature are frequently observed in the DECLIC experiments onboard the ISS, previous computational studies and their quantitative comparison to experimental results have focused exclusively on observations obtained in a velocity range where the solidification front is macroscopically flat. The phase-field simulations with the implementation of radial temperature gradients were shown to produce macroscopically curved solidification fronts that follow isotherms in steady state, and further demonstrated different evolution mechanisms of the primary spacing with a concave and a convex interface, respectively.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Articles in Other Journals or Periodicals	Ji K, Tabrizi AM, Karma A. "Isotropic finite-difference approximations for phase-field simulations of polycrystalline alloy solidification." Journal of Computational Physics. Submitted October 24, 2021. arXiv.org server: https://arxiv.org/abs/2110.12448 , Oct-2021

Articles in Peer-reviewed Journals	Dorari E, Ji K, Guillemot G, Gandin C-A, Karma A. "Growth competition between columnar dendritic grains - The role of microstructural length scales." Acta Materialia. 2022 Jan 15;223:117395. Article 117395. https://doi.org/10.1016/j.actamat.2021.117395 [Note: previously reported in September 2021 as in press; available online 13 October 2021.] , Jan-2022